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Nucleation, growth, percolation, and amorphous to crystalline transition of ultrathin molybdenum films

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We report on the nucleation, growth, percolation, and crystalline transition of ultrathin layers of molybdenum deposited on float glass substrate by *in situ* electrical properties. The transition from insulating to metallic state (*I-M*) is shown as a function of molybdenum film thickness and sheet resistance. In the island growth regime, the electrical conductance (G) has two exponential dependencies on thickness (t), namely, $\log G - B_1 \propto t$ and $\log G - B_2 \propto \sqrt{t}$, explained as the anisotropic and isotropic growth of islands, respectively. In the *I-M* transition region, the electrical conductance of the films follows the scaling law $G \propto (t - t_c)^q$. The value of critical exponent q determined from the present experiments agrees well with the theoretically predicted values for the critical exponent of conductivity in a two-dimensional percolating system. The amorphous to crystalline transition is also observed. The microscopic picture of film growth, derived from *in situ* sheet resistance measurements shows good agreement with that predicted by the percolation theory. © *2006 American Institute of Physics*. [DOI: 10.1063/1.2399325]

I. INTRODUCTION

Ultrathin films have been studied for decades to understand the nucleation and growth of metal islands that undergo coalescence resulting in a continuous percolating film during the early stages of the film growth.[1] The structural information on island films such as island density, island size, and fractional surface coverage are of great interest. These informations are necessary for understanding the kinetics of thin film growth. Recently, there has been interest in percolative characteristics of island films,[2] for example, the determination of the critical exponent q associated with percolation conductivity for the growth of thin metallic films on insulating substrates. The percolation theory has been discussed by Lobb and Frank[3] in two dimensions (2D) and by Kirkpatrick[4] in three dimensions (3D). Travenee and Markos[5] have theoretically calculated critical conductance distribution in various dimensions. Dubson and Garland[6] have used analog simulation techniques to study transition in electrical conductivity in 2D percolation systems. They have found that the critical exponent $q = 1.29 \pm 0.03$ for site percolation on a square lattice and $q = 1.34 \pm 0.07$ for a random void continuum percolation. The results are in agreement with theoretical predictions that the conductivity exponents for these two systems are the same. It has been predicted theoretically[7,8] that the critical exponent of conductivity q is, in general, nonuniversal, except in 2D growth, where it is universal. Han *et al.*[9] have fabricated a well-defined 2D anisotropic percolation system consisting of insulating random ellipsoids on a thin aluminum film. They have measured the critical exponent of conductivity q. They have found that q in an anisotropic percolation system approaches the same value as for the isotropic case near the critical region. They proved that the critical exponents should be independent of detailed microstructure as predicted by Lobb *et al.*[10] This universality of critical exponent has been observed experimentally by Octavio *et al.*[11]

The early stages in the growth of a metal film on an amorphous substrate consist of the nucleation of small metal clusters and then growth into islands. As the metal content increases, the islands grow in size and get interconnected to form infinite chains of metal. This enables continuous current paths extending through the material which "infill" to form continuous metal films at thickness $t = t_{\min}$. The electrical properties of such systems vary continuously as the metal content is changed. The conductance variation as a function of thickness is related to the fractional surface coverage with the growth of islands. The growth of islands is influenced by substrate temperature.[12] The percolation threshold t_c (also known as critical thickness) is considered as an insulator to metal (*I-M*) transition and is accompanied by a sharp drop in dc resistivity.[13] The effective thickness at which the resistance begins to fall depends on the type of substrates and substrate conditions, e.g., substrate temperature.[14] The mechanism of electrical conduction in discontinuous metal films on insulating substrate has been the subject of several publications including a review article by Morries and Coutts.[15] The main conduction mechanism is the electron tunneling between metal islands. Most theoretical treatments assume a homogeneous system. In practice, ultrathin evaporated metal films on insulating substrate are good systems for the study of the conduction mechanism in isolated metal particles. Particles of almost any desired size can be prepared by simply changing the experimental parameters such as average film thickness and the substrate temperature. The uniform deposition over the substrate is obtained by keeping a

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large source to substrate distance. The particle size is reasonably uniform in extremely thin[16] evaporated films deposited at substrate temperature ≤300 K.

The microstructure of the percolating system has been studied theoretically in terms of the island density and island size distribution by Amar *et al.*[17] They have presented the anisotropic island growth at low temperatures. Bruschi *et al.*[18] have studied the conformity of anisotropic island structure at low temperatures. They have studied the temperature dependence of the nucleation, growth, and percolation of thin metal film growth on amorphous substrate using Monte Carlo simulations. They observed that the size and shape of islands change with temperature. They have also shown that the transition temperature from anisotropic island to isotropic island occurs in the temperature interval of 37–200 K. The microstructure of the percolating system has also been studied experimentally, such as, for example, gold films using transmission electron microscopy (TEM),[19] indium oxide films using *in situ* conductivity,[12] and antimony films using both TEM and *in situ* conductivity methods.[20]

The determination of insulator to metallic transition region and fractional surface coverage and hence critical coverage using electron microscopy may be erroneous due to the contamination of ultrathin film when exposed to air. *In situ* resistance measurement is a sensitive method to study the percolating system experimentally. This method has been applied for various thin films, such as, for example, silver,[11] indium oxide,[12] platinum,[14] antimony,[20] nickel,[21] chromium,[22] and bismuth films.[23] One of the difficulties in the experimental determination of the critical exponent of conductivity is the determination of the critical fractional coverage x_c and the *I-M* transition region. The *I-M* transition region and the critical exponent of conductivity have been determined by the least-square fit method[21,22] of *in situ* resistance data.

Here we report a simple method for the experimental determination of *I-M* transition region and the critical exponent of electrical conductivity in the percolating system. The method presented in the present paper is applied to molybdenum (Mo) films. Mo is an important material for large-scale integration/very large-scale integration metallization because of its low resistivity and high melting point.[24] Mo is also used as a back contact for advanced solar cell assemblies.[25] Alternate layers of Mo with low-Z spacer elements such as silicon, carbon, beryllium, and boron carbide are used as x-ray multilayer mirrors.[26–29] These mirrors are finding extensive applications as optical elements in extreme ultraviolet lithography, plasma diagnostics, spectroscopy, and astronomy. For a real multilayer to exhibit the theoretical reflectivity, it is essential that Mo layers are continuous with sharply defined boundaries. This requires the film growth mechanism to be predominantly two dimensional. Minimizing absorption at soft x-ray wavelengths would enhance reflectivity. This requires the thickness of Mo layers to be just above the $t_{\min}$ for which the film is continuous.[30,31] A clear understanding of the nucleation and growth mechanisms of Mo layers in such structures would allow the deposition of optimal thicknesses. It is also important to investigate growth mode and discontinuous to continuous transition in ultrathin Mo films. We also report the observation of amorphous to crystalline transition for Mo films. We have determined the structural parameters such as the relative average nucleation density, island area, and fractional surface coverage of Mo films. The microscopic picture of island film growth derived from *in situ* sheet resistance measurements shows a good agreement with percolation theory. We also provide an experimental evidence for the universality of the critical exponents of conductivity in a 2D percolating system.

II. EXPERIMENTAL DETAILS

Mo films were deposited on smooth (rms surface roughness ~3 Å as determined using hard x-ray reflectivity) float glass substrates at 300 and 100 K using an ultrahigh vacuum (UHV) electron beam evaporation system. The substrate was cooled by liquid nitrogen, and the temperature was measured by a thermocouple. The base pressure in the chamber during deposition was $\sim 5.9\times 10^{-7}$ Pa for substrate at 300 K (film 1) and $\sim 1.2\times 10^{-7}$ Pa at 100 K temperature (film 2), respectively. The source to sample distance was kept at ~0.5 m to ensure uniform deposition. The purity of molybdenum used was 99.9%. A quartz crystal oscillator, coupled to programable frequency counter/timer, measured the thickness of the film in each run. The frequency counter was coupled to a personal computer through RS232 interface. The quartz crystal was water cooled to minimize the thermal drift in the quartz crystal. A mask was installed in front of substrate to reduce the thickness error. The deposition rate was kept at 0.1 Å/min and was monitored using quartz crystal.

There may be two types of errors in the measurements of thickness: (a) random errors and (b) relative errors due to thermal drift in quartz crystal. The random errors were estimated to be within ~0.7% by repeating the experiment eight times. The relative errors were obtained by depositing alternating layers of Mo and Si with varying Mo thickness from 10 to 50 Å with a periodicity of 100 Å and ten layer pairs. The inner layers do not get contaminated when exposed to ambient. The individual deposited layer thicknesses were checked by hard x-ray reflectivity measurement using Cu $K\alpha$ wavelength. A relative error of ~0.2% was obtained. The total error in the measurement of thickness is 0.7%.

The thickness dependence of film sheet resistance (R_s) was measured *in situ* using Van der Pauw pressure contact method. The Ohmic contacts to the predeposited gold film (~1000 Å) at four points on the substrate were made using gold tips with diameter ~3 mm. The electrodes were spaced 20 mm apart. The sheet resistance during the growth of the film was measured by passing current from a current source having accuracy and stability of ±2%. Voltage was measured using a digital multimeter with a minimum accuracy of ±10 μV and a maximum error of 50 μV.

III. RESULTS AND DISCUSSIONS

Figure 1 shows the sheet resistance as a function of Mo film thickness for films 1 and 2. Four distinct regions are clearly discernible for both the films (inset of Fig. 1). For example, for film 1 (300 K), an initial thickness region of $t \approx 0.01-0.35$ nm indicated as A in the figure, the sheet resis-

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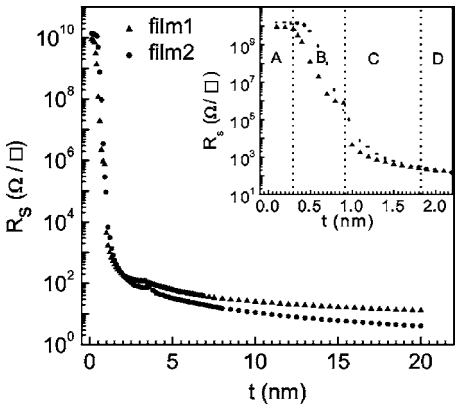

FIG. 1. The sheet resistance R_s vs thickness t for Mo thin films. The solid up triangles (▲) are for film 1 deposited at 300 K and the solid circles (●) are for film 2 deposited at 100 K. The inset shows four different regions of film growth.

tance R_s has no significant change and R_s is predominantly that of the substrate surface. In the thickness interval of $t \approx 0.35-0.9$ nm indicated as B, R_s decreases with increasing thickness significantly. In the third region $t \approx 0.9-1.8$ nm indicated as C, R_s decreases over several orders of magnitude with increasing thickness. In the fourth region $\sim t > 1.8$ nm marked as D, R_s decreases very slowly with increasing thickness. The resistivity drops from 1×10^7 to 2.42 $\mu\Omega$ m and from 5×10^7 to 0.37 $\mu\Omega$ m for films 1 and 2, respectively. For better clarity, the inset in Fig. 1 shows the experimental data of Mo films up to a thickness of 2.2 nm. The experimental points displaying distinct concave upward trend and downward feature at thicknesses $t \approx 0.9$ and 1.1 nm indicate the existence of critical points for films 1 and 2, respectively. We discuss these distinct regions of film growth briefly in the subsequent sections.

A. Nucleation and growth

For the convenience of data analysis, the experimental data are plotted in terms of conductance versus thickness in Fig. 2. The region (A), before the onset of the conductance, can be attributed the nucleation stage. The onset of conductance occurs early for film 1 compared to film 2 indicating more nucleation density in film 2 compared to film 1 (discussed in next paragraph). Different nucleation densities for these two different deposition conditions are expected, as the nucleation rate depends on the ability of the adsorbed atoms to diffuse and collide with each other.[32] The activation energy for surface diffusion is small for film 2. This may be due to low deposition temperature. This implies that the nuclei grow only from material received by direct impingement from vapor phase and with higher binding energy of adsorbed single atom to the substrate.

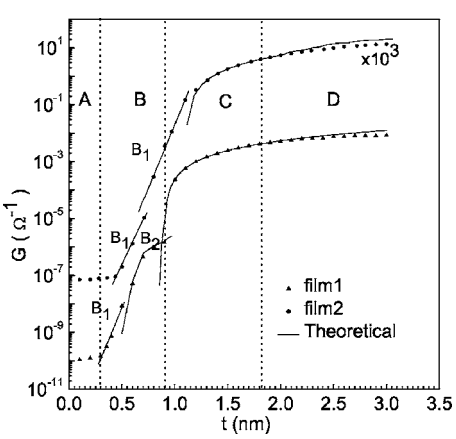

FIG. 2. The conductance G vs t plots of sheet resistance measurements. The solid up triangles (▲) are for film 1, and solid circles (●) are for film 2.

In our present studies on ultrathin films, we assume that islands are reasonably uniform in size.[16] Large source to substrate distance ensures uniform deposition over the substrate. In a homogeneous system, the electronic conduction in the discontinuous metal films on the insulating substrate takes place through tunneling from one metal island to the neighboring island due to the overlap of the electron wave functions at the neighboring islands. According to this model, the film conductance increases exponentially as the interisland distance shrinks,[12,33]

$$G = G_o e^{-\alpha r}, \tag{1}$$

where α^{-1} is the characteristic tunneling distance of the electron wave function, r is the distance between the islands defined as $r = l - \beta$. Here, l is the average distance between the island centers and β is the characteristic linear dimension of the island. The number of the islands N per unit area is inversely proportional to l^2. Using Eq. (1), the growth region (region B in Fig. 2) is best fitted by two exponential dependences of G on t. For film 1, $\log G - B_1 \propto t$ up to a thickness of 0.5 nm and $\log G - B_2 \propto \sqrt{t}$ up to a thickness of 0.9 nm. For film 2, $\log G - B_1 \propto t$ up to a thickness of 1.1 nm. The dependences $\log G - B_1 \propto t$ and $\log G - B_2 \propto \sqrt{t}$ have been attributed[12] to the anisotropic and isotropic growths of islands, respectively. The term anisotropic applies to islands of irregular shape, and the term isotropic is used for nearly circular islands. The microscopic picture changes from anisotropic to isotropic, at a thickness of 0.5 nm, for film 1 due to the coalescence of islands. The coalescence process occurs for film 2 at a thickness of 0.7 nm and is indicated by a sudden slope change during island growth (Fig. 2). The average area and the density of islands are calculated from the slope of the straight line before the coalescence of islands (the height of isolated islands is relatively constant assuming 2D film growth). The ratio of average island area between the films 1 and 2 is 1.15 and the ratio of average island density is 0.86. This shows that the islands are more compact and small in size during the growth of film 2 as compared to film 1. The low density of nucleation is shown by the early onset of conductance for film 1 as compared to film 2 (Fig. 2). During the coalescence process, the transition from anisotropic to isotropic growth occurs for film 1, while for film 2 growth mode does not change. The change in island shapes during coalescence is due to the driving force[34] for film 1. The driving force arises from the thermal energy at the substrate surface. At low temperature, the island growth occurs by the direct impingement of adatoms, resulting in anisotropic islands, while at high temperature the island growth occurs by the surface diffusion of adatoms. The large islands formed by the incorporation of small mobile islands behave

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in a liquidlike manner to minimize surface free energy, resulting in isotropic islands. At a particular substrate temperature, the variation of conductance with film thickness is due to variation in the distribution of islands. As the substrate temperature changes, the nature of nucleation density and the growth mode of island also change, as we have observed for films 1 and 2. A similar microstructure of film growth has been observed experimentally for other films. Examples are the isotropic growth of gold films[19] and the isotropic and anisotropic growths of indium oxide films.[12] The microscopic picture of Mo film growth obtained from *in situ* sheet resistance measurements is consistent with the theoretical simulation study of Bruschi *et al.*[18] for the isotropic and anisotropic growths of thin metal films on amorphous substrate in the percolating system. Our observation of transition in growth mode, i.e., the isotropic and anisotropic growths of Mo, is similar to scanning tunneling microscopy studies of Ag films of Caspersen *et al.*[35] They have observed the existence of a transition in growth mode with different substrate temperatures.

B. Study of infinite cluster: An insulator-metal transition

In the percolating stage or the *I-M* transition region (marked as C in Fig. 2), large islands that are formed during the coalescence process grow further and join to form an infinite cluster of metallic network structure. As the deposition continues, secondary nucleation occurs in these channels. The secondary islands grow and touch the sides of these channels. At the same time, the channels are bridged and filled rapidly in a liquidlike manner. As the deposition continues, most of the channels are eliminated but still contain many small irregular holes.[32] The secondary nucleation takes place on the substrate within these holes, and the growing nuclei are incorporated in a liquidlike manner to form a continuous film. The thickness at which the infinite cluster is formed is called the critical thickness t_c. The minimum film thickness below which film is discontinuous is identified as $t_{\min}$. The film undergoes *I-M* transition in the region $t_c < t < t_{\min}$. In the *I-M* transition region, the conductance of the film is described by the scaling law,[4,36,37]

$$\sigma(x) \propto (x_c - x)^{-q} \quad \text{when } x < x_c \tag{2a}$$

$$\propto (x - x_c)^{q}, \quad x > x_c, \tag{2b}$$

where x is the occupation probability of a conducting area on the insulating substrate (also known as fractional coverage), x_c is the critical value of fractional coverage, and q is the critical exponent of the electrical conductivity. The value of q is a consequence of the film growth mode. With the assumption[21] that $(x-x_c)$ is proportional to $(t-t_c)$ near x_c, the conductance should scale as $(t-t_c)$.

The upper boundary limit point of the *I-M* transition region marks the onset of continuous film at the thickness $t_{\min}$. The thickness $t_{\min}$ is identified from the minimum in the $R_s t^2$ vs t graph shown in Fig. 3. The $R_s t^2$ vs t graph has been discussed in more detail in Sec. III C for the calculation of average grain diameter. The inset in Fig. 3 shows the derivative trace, i.e., $(d/dt)(R_s t^2)$ vs t. The $t_{\min}$ obtained from de-

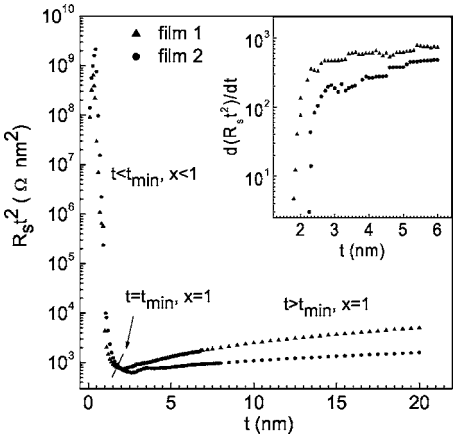

FIG. 3. The $R_s t^2$ vs t plot of sheet resistance measurements. Solid up triangles (▲) are for film 1, and solid circles (●) are for film 2. The inset shows the derivative trace, i.e., $(d/dt)(R_s t^2)$ vs t.

rivative trace near $t=t_{\min}$ is 1.8 and 2.2 nm for films 1 and 2, respectively. Rycroft and Evans[38] have identified $t_{\min}$ from the minimum in the Rt^2 vs t plot for Fe, Pt, and Cu films. The higher $t_{\min}$ value for film 2 is due to low surface mobility. We have discussed it in more detail by correlating it with x-ray reflectivity analysis.[39] Veldkamp *et al.*[31] sputter deposited Mo films on sapphire substrate and obtained $t_{\min}=2.5$ nm at 300 K, which is different from $t_{\min}=1.8$ nm in the present case. The difference in $t_{\min}$ value is due to different substrates used and/or different deposition techniques adopted which may give different surface mobilities of Mo atoms. The adhesion of deposited films to the substrate is strongly dependent on the chemical nature, microscopic topography, and cleanliness of the substrate surface.[40] We deposited Mo on float glass substrate, and Veldkamp *et al.* have deposited on sapphire, resulting in differences in $t_{\min}$ value due to different surface mobilities. Again, atomic clusters of Mo may be incorporated in sputter deposited films as adopted by Veldkamp *et al.*, which in turn decrease surface mobilities. Thus the sputter deposited film would require more atoms to be deposited to form continuous films and hence higher $t_{\min}$ value.

The conduction mechanism is dominated by tunneling below t_c as discussed earlier. At $t \geqslant t_c$, the metallic conduction starts contributing to the conduction mechanism. The experimental deviation of tunneling current occurs at thicknesses $t_c \approx 0.9$ and 1.1 nm for films 1 and 2, respectively (Fig. 2). To obtain q and actual values of t_c, Eq. (2b) is used with t_c value as a starting guess. Figure 4 shows the variation of G against $t-t_c$ for different values of t_c. Despite the variation of t_c in the narrow range, only the curves with critical thicknesses $t_c=0.9$ and 1.1 nm come close to a straight line for films 1 and 2, respectively. A least-square fit to this curve yields critical exponents $q=1.29\pm0.02$ and 1.3 ± 0.02 for films 1 and 2, respectively. We have obtained similar data for eight separate evaporations that yield averages for $t_c=0.9\pm0.01$ and 1.1 ± 0.02 nm with critical exponents 1.29 ± 0.02 and 1.3 ± 0.02 for films 1 and 2, respectively. The slope change in Fig. 4 is due to amorphous to crystalline phase transition in the films and has been discussed in more detail in Sec. III C. The value of the critical conductivity

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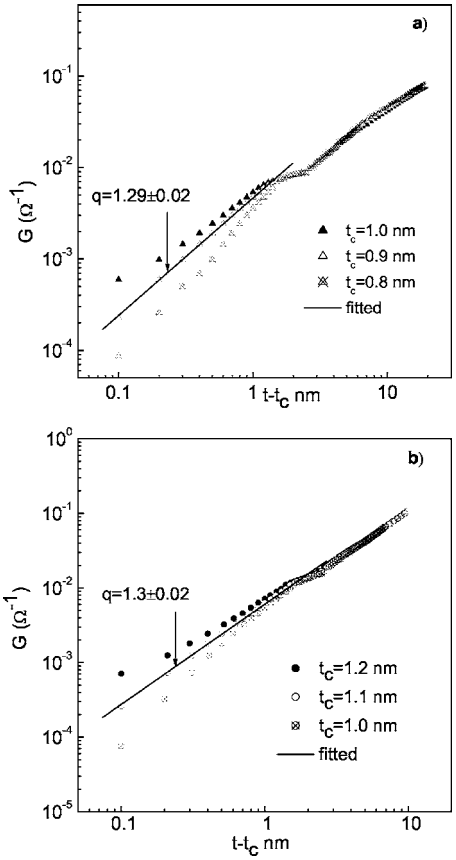

FIG. 4. The G vs $t-t_c$ plot for three different values of t_c as indicated in the figure (a) for film 1 and (b) for film 2. The solid line corresponds to the linear fit of the points for which $t-t_c<1.6$ and 2.4 nm for films 1 and 2, respectively, where the island growth should be still horizontal.

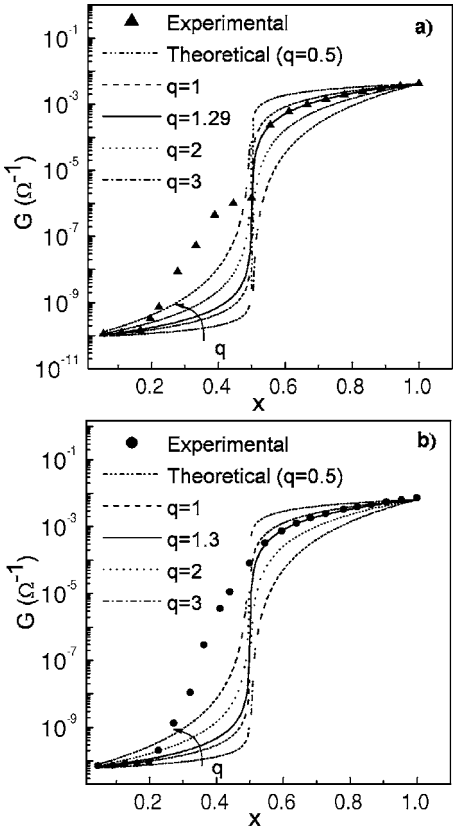

FIG. 5. The G vs fractional coverage x. (a) Solid up triangles (▲) are for film 1, and (b) solid circles (●) are for film 2. The dotted and solid lines are obtained from best fit to Eq. (2) for different values of q as indicated in the figure. The solid line indicates the best-fit data of $q=1.29\pm0.02$ and 1.3 ± 0.02 for films 1 and 2, respectively.

exponent agrees within the experimental error with the 2D theoretical model calculations of Dubson and Garland (Ref. 6). The error of 0.01 in t_c and q is because of the overlapping of different conduction mechanisms at both ends of the *I-M* region. Suresh[41] has plotted sheet resistance versus thickness for molybdenum thin films at 300 K. We have calculated the critical exponent of conductivity by replotting his data in this method. The critical exponent of conductivity obtained from his data is $q=1.3\pm0.02$ which is consistent with our experimental results.

The critical fractional coverage x_c is obtained from the proportionality constant in Eq. (2). In 2D growth the fractional coverage is related to thickness[21,38] as $x=kt$. The parameter k is obtained with the boundary conditions,[38] at $t=0$, $G=G_D$ and at $t=t_{\min}$, $G=G_M$. G_D and G_M are the conductance of dielectric substrate and conductance at the onset of continuous film, respectively. The ordinate intercept in Fig. 4 yields the parameters $k=0.55$ and 0.45 nm^{-1} for films 1 and 2, respectively. The experimental G_M values are taken for calculation of k, $G_M=4.1\times10^{-3}$ and $8.3\times10^{-3}\Omega^{-1}$ for films 1 and 2, respectively. The critical fractional coverage obtained is $x_c=kt_c=0.5$ for both the films. This agrees well with 2D film growth.[13] Figure 5 shows conductance against fractional coverage. The experimental data are theoretically simulated using Eq. (2) for different values of q. The best fit is obtained for $q=1.29$ and 1.3 for films 1 and 2, respectively. The simulation does not fit below critical thickness for films 1 and 2 as shown in Figs. 5(a) and 5(b), respectively. This implies that for our case the relation $\sigma(x)\propto(x_c-x)^{-q}$ is not valid below the critical thickness. The discrepancy may be due to the electron tunneling between particles, which plays an important role when metal islands and the separations are small (for high melting temperature metals[14]).

We show the comparison of our results with the least-square method used earlier by various authors[21,22] for the calculation of the *I-M* transition region and the critical exponent of conductivity. Sliding least-square fit using Eq. (2) is done on 3.5 nm segments of experimental data (Fig. 1) with starting thickness t_s specified for each segment. For each value of t_c and q, the corresponding χ^2 value is obtained. Results of least-square fit are shown in Figs. 6(a) and 6(b) for film 1 and 2, respectively. The χ^2 value is minimum for $t_c=0.9\pm0.01$, $q=1.29\pm0.02$ for film 1 and $t_c=1.1\pm0.02$, $q=1.3\pm0.02$ for film 2. t_c and q values are nearly constant in the starting thickness ranges of $0.9\leqslant t_s\leqslant1.8$ and $1.1\leqslant t_s\leqslant2.2$ nm for films 1 and 2, respectively. The same procedure is repeated for the segment widths of 3, 4, 4.5, and 5.5 nm, giving identical results, implying that the *I-M* transition region and q values obtained by two different methods are the same. The confidence in the experimental values of t_c, $t_{\min}$, and q is generated by the repetition of experiments over eight separate runs. The data are reproducible from run to run, yielding consistent data. Although the critical thickness is different, the critical exponents of conductivity are identical

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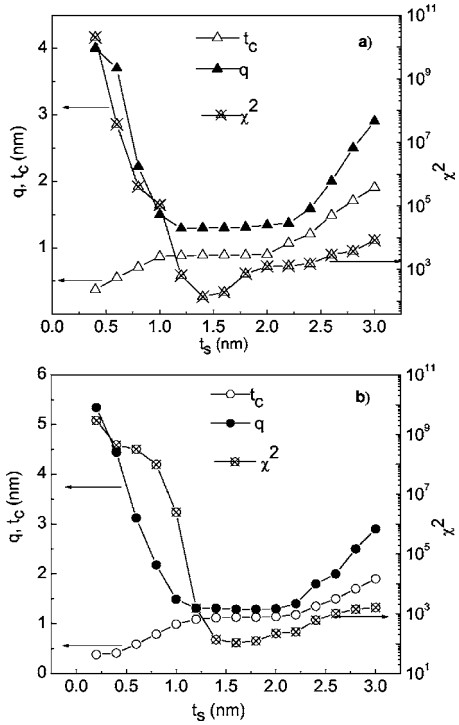

FIG. 6. The critical thickness t_c, the critical exponent of conductivity q, and χ^2 vs the starting thickness t_s obtained from least-square fit method (a) for film 1 and (b) for film 2.

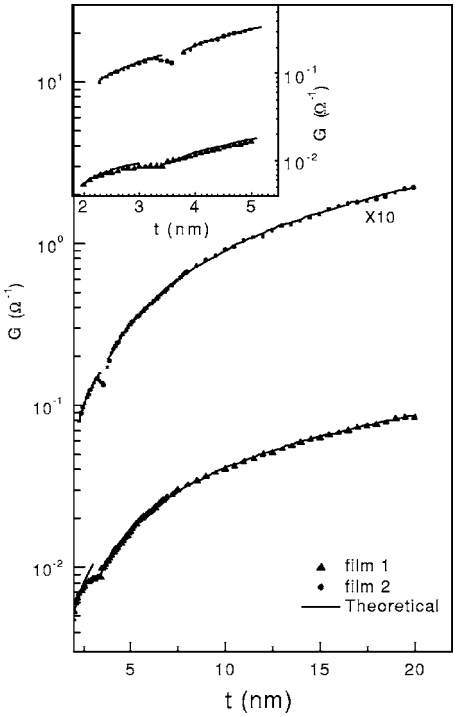

FIG. 7. The G vs t measurements in the region D ($t > t_{\min.}$). The solid up triangles (▲) are for film 1, and solid circles (●) are for film 2. The solid line is obtained from best fit to Eq. (3). The inset shows clear visibility of conduction jump at thicknesses t=2.5 and 3.3 nm in films 1 and 2, respectively.

within experimental error in two films with different microscopic pictures. This provides evidence of the existence of universality of critical exponents of conductivity in 2D percolating system.

C. Amorphous to crystalline transition

In the thickness region $t > t_{\min}$, the scattering of electrons by thin film surfaces and grain boundaries are used to model the continuous film conductivity. The film sheet resistance in the metallic continuous region D (Fig. 1) has been fitted using the relation,[42]

$$Rt = A_0 + A_1/t + A_3/t^3, \tag{3}$$

where the second and third terms, contribute to conduction due to grain boundary and roughness scattering, respectively. The fitted plots of metallic regions ($t > t_{\min}$) are shown in Fig. 7. The best-fit results are tabulated in Table I. It is clear that the grain boundary contribution (A_1) is zero up to thicknesses of 2.5 and 3.3 nm for films 1 and 2, respectively. This implies that the films are amorphous up to the above mentioned thickness. The nonzero A_1 values for thicknesses greater than 2.5 and 3.3 nm for films 1 and 2, respectively, imply that there is a contribution of grain boundaries to the conductance in the films. The films undergo amorphous to crystalline transition at thicknesses of 2.5 and 3.3 nm for films 1 and 2, respectively. This transition can be visualized from the cusp observed in the conduction (inset in Fig. 7). This amorphous to crystalline transition is the cause of slope change in Figs. 4(a) and 4(b) for films 1 and 2, respectively. A similar amorphous to crystalline transition has been observed for antimony films (Ref. 20). In Table I, the A_1 value is more for film 1 compared to film 2 indicating that grain boundary scattering is more in film 1. The *in situ* sheet resistance data are used to calculate the average grain diameter D_g. The average grain diameter is related to resistivity as[38] $D_g = \rho_0 \lambda_0 (\rho_\infty - \rho_0)^{-1}$, where λ_0 is electron mean free path in the material and is given as $\lambda_0 = hk_f / 2\Pi n e^2 \rho_0$. Here k_f is the wave vector at the Fermi surface, n is the electron density, e is the electron charge, ρ_∞ is the resistivity of an infinitely thick film, and ρ_0 is the resistivity of the bulk material. For convenience to analyses in terms of size effects,[43] the measured sheet resistance data have been plotted in terms of $R_s t^2$ as a function of film thickness in Fig. 3. The film resistivity is linearized when $R_s t^2$ is plotted with film thickness for $t \gg t_{\min}$. The slope of the linear region yields a physical parameter ρ_∞, whereas an ordinate intercept yields the specular parameter. The values of ρ_∞ are 0.242 and 0.037 $\mu\Omega$ m for film 1 and 2, respectively. The values of ρ_0 are 5.7×10^{-2} and $9.2 \times 10^{-3} \mu\Omega$ m at 300 and 100 K, respectively.[44] Thus, the average grain diameters D_g obtained for films 1 and 2 are 3.1

TABLE I. The best-fit results to Eq. (3) obtained in the continuous region $t > t_{\min}$ for films 1 and 2.

	Film 1 thickness (nm) range		Film 2 thickness (nm) range	
Parameter	1.8–2.5	2.5–20	2.2–3.3	3.3–20
A_0 (Ω nm)	298.06	223.34	188.06	73.3
A_1 (Ω nm^2)	0	223.34	0	208.09
A_3 (Ω nm^4)	598.73	1583.68	1093.19	3846.74

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and 6.5 nm, respectively (assuming bcc Mo and Fermi surfaces with spherical shape). The roughness scattering is more for film 2 compared to film 1. This is reflected as the higher A_3 for film 2 compared to film 1 (Table I). The more roughness scattering for film 2 is due to low atomic surface mobility, which in turn increases rms surface roughness. We have also checked the density of our films using x-ray reflectivity.[39] The density obtained from reflectivity are 99% and 97% of bulk density for films 1 and 2, respectively. The high density shows low concentrations of primary voids in our films and reflects the good quality of Mo ultrathin films.

IV. CONCLUSION

We have given a direct experimental method to determine the *I-M* transition region and the critical exponent of electrical conductivity for a percolation system taking Mo ultrathin films as an example. The upper limit of *I-M* transition is obtained from the minimum in the $R_s t^2$ vs t plot. The deviation of conduction mechanism from tunneling gives the lower limit. The critical exponents of conductivity are $q = 1.29 \pm 0.02$ and 1.3 ± 0.02, with critical thicknesses $t_c = 0.9 \pm 0.01$ and 1.1 ± 0.02 nm for Mo films deposited at 300 and 100 K, respectively. The values agree well with those obtained by using a least-square fit method. The critical fractional coverage x_c is 0.5 for both the films. The island growth is isotropic and anisotropic for Mo films deposited at 300 and 100 K, respectively, after coalescence process. The information on relative average nucleation density, island area, and fractional surface coverage are derived from *in situ* sheet resistance measurements. This makes possible the physical interpretation of percolating system. The study shows that Mo films grown on float glass substrate are predominantly 2D. Mo films deposited at two different temperatures show different microscopic pictures, but the critical exponents of conductivity have identical values. This provides evidence for the universality of conductivity exponents in 2D percolating system. We also identified the crystalline transition in Mo films from the conductivity jump in the films.

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